fn sample_discrete_gaussian

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn sample_discrete_gaussian` in mod.rs at commit 0be3ab3e6 (outdated[1]). This proof is an adaptation of subsection 5.3 of [CKS20].

1 Vetting history

- Pull Request #519

2 Pseudocode

Precondition

$\texttt{scale} \in \mathbb{Q} \wedge \texttt{scale} \geq 0$

Implementation

```
def sample_discrete_gaussian(scale) -> int:
    if scale == 0:
        return 0

    t = floor(scale) + 1
    sigma2 = scale**2

    while True:
        candidate = sample_discrete_laplace(t)
        x = abs(candidate) - sigma2 / t
        bias = x**2 / (2 * sigma2)
        if sample_bernoulli_exp(bias):
            return candidate
```

Postcondition

`sample_discrete_gaussian` returns an integer sample from $\mathcal{N}_{\mathbb{Z}}(0, scale^2)$, or an error propagated from `sample_discrete_laplace` or `sample_bernoulli_exp`.

[1] See new changes with `git diff 0be3ab3e6..d632d8a rust/src/traits/samplers/cks20/mod.rs`

3 Proof

First, define the target distribution.

Definition 3.1. (Discrete Gaussian). Let $\mu, \sigma \in \mathbb{R}$ with $\sigma > 0$. The discrete gaussian distribution with location μ and scale σ is denoted $\mathcal{N}_{\mathbb{Z}}(\mu, \sigma^2)$. It is a probability distribution supported on the integers and defined by [CKS20]

$$\forall x \in \mathbb{Z} \quad P[X = x] = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sum_{y \in \mathbb{Z}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}} \quad \text{where } X \sim \mathcal{N}_{\mathbb{Z}}(\mu, \sigma^2)$$

We must show that the return value of `sample_discrete_gaussian(scale)`, conditioned on not returning an error, is a sample from $\mathcal{N}_{\mathbb{Z}}(0, scale^2)$.

Let $t = \lfloor \sigma \rfloor + 1$. Now fix any iteration of the loop.

Lemma 3.2. If y is a realization of $Y \sim \mathcal{L}_{\mathbb{Z}}(0, \sigma)$, and c is a realization of $C \sim Bernoulli(exp(-(|y| - \sigma^2/t)^2/(2\sigma^2)))$, then $E[C] = \frac{1-e^{-1/\sigma}}{1+e^{-1/\sigma}} e^{-\frac{\sigma^2}{2t^2}} \sum_{y \in \mathbb{Z}} e^{-\frac{y^2}{2\sigma^2}}$ [CKS20].

Proof.

$$\begin{aligned}
E[C] &= E[E[C|Y]] \\
&= E[e^{-\frac{(|Y|-\sigma^2/t)^2}{2\sigma^2}}] && \text{since } E[Bernoulli(p)] = p \\
&= \frac{1-e^{-1/\sigma}}{1+e^{-1/\sigma}} \sum_{y \in \mathbb{Z}} e^{-\frac{(|y|-\sigma^2/t)^2}{2\sigma^2} - |y|/t} && \text{expectation over } Y \sim \mathcal{L}_{\mathcal{Z}}(0, \sigma) \\
&= \frac{1-e^{-1/\sigma}}{1+e^{-1/\sigma}} e^{-\frac{\sigma^2}{2t^2}} \sum_{y \in \mathbb{Z}} e^{-\frac{y^2}{2\sigma^2}}
\end{aligned}$$

□

Theorem 3.3. If y is a realization of $Y \sim \mathcal{L}_{\mathbb{Z}}(0, \sigma)$ and c is a realization of $C \sim Bernoulli(exp(-(|y| - \sigma^2/t)^2/(2\sigma^2)))$, then $P[Y = y | C = \top] = \frac{e^{-\frac{y^2}{2\sigma^2}}}{\sum_{y' \in \mathbb{Z}} e^{-\frac{y'^2}{2\sigma^2}}}$. That is, $Y|_{C=\top} \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ [CKS20].

Proof.

$$\begin{aligned}
P[Y = y | C = \top] &= \frac{P[C = \top | Y = y] P[Y = y]}{P[C = \top]} && \text{Bayes' Theorem} \\
&= \frac{e^{-\frac{(|y|-\sigma^2/t)^2}{2\sigma^2}} \frac{1-e^{-1/t}}{1+e^{-1/t}} e^{-|y|/t}}{E[C]} && \text{by definition of } \mathcal{L}_{\mathbb{Z}}(0, \sigma) \\
&= \frac{e^{-\frac{(|y|-\sigma^2/t)^2}{2\sigma^2}} e^{-|y|/t}}{e^{-(\sigma/t)^2/2} \sum_{y' \in \mathbb{Z}} e^{-\frac{y'^2}{2\sigma^2}}} \\
&= \frac{e^{-\frac{y^2}{2\sigma^2}}}{\sum_{y' \in \mathbb{Z}} e^{-\frac{y'^2}{2\sigma^2}}}
\end{aligned}$$

□

Since the function's return value is a draw from $Y \sim \mathcal{L}_{\mathbb{Z}}(0, scale)$ conditioned on $C = \top$, then by 3.3, the return is distributed $\mathcal{N}_{\mathbb{Z}}(0, scale^2)$.

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.